

NAVEEN KUDLI BALAJI

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Experience

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Jul 2023 – Jun 2025

Associate Software Engineer

Rochester, NY

- Developed and integrated a microservice to ensure data integrity across legacy and Salesforce systems using Java, Kafka, C# and Oracle SQL. Implemented contact reload feature for returning clients.
- Delivered 500k in annual cost reductions by automating worker migration in Paychex Flex using microservices, to streamline business processes.
- Designed and developed REST APIs to enhance robustness and functionality of legacy system to facilitate updating existing records.
- Improved UI functionality of the system to accurately reflect user workflow status.
- Automated code deployment process to dev servers, boosting productivity by 20%.
- Facilitated technical deep dive sessions for a team of 6 developers, improving Agile development processes and boosting team productivity.

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Jan 2022 – Aug 2022

Software Development Engineer Co-op

Rochester, NY

- Investigated and implemented external API integration as part of end-to-end workflow in Paychex Flex.
- Developed a use-case to evaluate integration of Kafka consumer into the existing data pipeline.
- Created dynamic Angular components for Paychex Flex, and optimized MongoDB environments.

Infrd.ai

Jul 2019 – Jun 2020

Software Engineer

Bangalore, India

- Engineered a robust file upload module, enabling comprehensive support for various file formats and streamlining data ingestion.
- Integrated machine learning and backend pipelines enabling users to train and fine-tune custom models.
- Enhanced user dashboard experience by implementing new features, including document annotation and pagination-based loading.
- Designed and developed CRUD APIs for information retrieval from Elasticsearch.

Technical Skills

Languages: Python, Java, C#, C++, Angular

Tools and Frameworks: Git, Springboot, Splunk, Openshift, Pytorch, AWS,

Education

Rochester Institute of Technology

M.S Computer Science

Rochester, NY

BNM Institute of Technology

B.E Computer Science

Bangalore, India

Projects

Confusion Estimation using Language and Speech

May 2023 - Jul 2023

- Developed a multimodal machine learning system as part of my capstone to estimate user confusion by integrating speech and language features, improving detection accuracy by over 5% compared to unimodal baselines.
- Engineered and fused features from the MULTICOLLAB dataset, extracting Mel-Frequency Cepstral Coefficients (MFCCs) from audio signals and processing text transcripts to create a unified feature set for classification.
- Trained, evaluated, and compared multiple classifiers (Random Forest, LSTM, Decision Tree) to establish performance baselines and demonstrate the effectiveness of an early-fusion multimodal approach.
- Performed exploratory data analysis to identify key behavioral indicators of confusion, such as increased utterance length and word repetition, and rigorously evaluated final model performance using precision, recall, and F1-scores.

Emotion Recognition from Speech

Nov 2021 – Dec 2021

- Developed a system to detect users sentiment based on speech input using Mel Frequency Cepstral Coefficients (MFCCs) with an accuracy of **83%**.